# 5.5 Interpolation, Quotient Rings, and the Chinese Remainder Theorem

Section 5.4 proved a uniqueness statement: two polynomials of degree at most $n$ that agree at $n+1$ distinct points are equal (Theorem 5.4.2 (a)). It said nothing about existence. Is there, for instance, a quadratic through the points $(0,1)$, $(1,3)$, $(2,7)$? This section builds one by a formula. It then reads the same question as three congruences. By Theorem 5.1.3 (a), "the value at $c$ is $y$" says that $f$ leaves the remainder $y$ on division by $x-c$. So we look for $f$ with

$$f \equiv 1 \pmod{x}, \qquad f \equiv 3 \pmod{x-1}, \qquad f \equiv 7 \pmod{x-2}.$$

Three congruences to coprime moduli: that is the Chinese remainder theorem of *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapter 3 (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.3), with polynomials in place of integers. On the way we meet the classes modulo a polynomial $m$. They form a ring $F[x]/\langle m\rangle$, and it is a field exactly when $m$ is irreducible, as $\mathbb{Z}/p\mathbb{Z}$ is a field exactly when $p$ is prime.

Throughout, $F$ is a field. "Polynomial" means an element of $F[x]$ unless another field is named.

## 5.5.1 Polynomials through given points

The points at which values are prescribed are called **nodes**. Suppose the nodes are $c_0,\dots,c_n \in F$ and the values $y_0,\dots,y_n \in F$. Fitting all the values at once is hard. Fitting one value while ignoring the others would be easy if we had, for each node $c_i$, a polynomial that is 1 at $c_i$ and 0 at every other node: then $y_0$ times the first, plus $y_1$ times the second, and so on, has every value right. A polynomial that vanishes at the nodes $c_j$ with $j \neq i$ is a multiple of the product of the factors $x-c_j$ (Theorem 5.4.1); the simplest is that product itself. Its value at $c_i$ is the product of the numbers $c_i - c_j$, and dividing by that product makes the value 1.

*Each Lagrange polynomial is* 1 *at its own node and* 0 *at all the others.*

**Definition 5.5.1. Lagrange Polynomials**

Let $n \in \mathbb{N}$ and let $c_0,\dots,c_n \in F$ be **distinct**. For $i \in \{0,\dots,n\}$, the $i$-th **Lagrange polynomial** of these nodes is

$$\ell_i := \prod_{\substack{j=0 \\ j\neq i}}^{n} (c_i - c_j)^{-1}(x - c_j) \in F[x],$$

the product, in increasing order of $j$, of the $n$ polynomials $(c_i-c_j)^{-1}(x-c_j)$ with $j \neq i$.

The inverse $(c_i-c_j)^{-1}$ exists because $c_i \neq c_j$ for $j \neq i$, and $F$ is a field. The product of a list of polynomials is given by the recursion of Sets, Relations, and Cardinality, Definition 2.3.1, read in the commutative ring $F[x]$ (Theorem 4.2.4 (a)); the pattern *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$ used for lists of integers.

**Examples.**

- *Two nodes.* For the nodes $0,1 \in \mathbb{Q}$: $\ell_0 = (0-1)^{-1}(x-1) = 1-x$ and $\ell_1 = (1-0)^{-1}(x-0) = x$. Indeed $\ell_0(0)=1$, $\ell_0(1)=0$, $\ell_1(0)=0$, $\ell_1(1)=1$.

- *Three nodes.* For the nodes $0, 1, 2 \in \mathbb{Q}$, the polynomial $\ell_0$ is
$$\ell_0 = (0-1)^{-1}(x-1)\cdot(0-2)^{-1}(x-2) = \tfrac{1}{2}(x-1)(x-2).$$
It vanishes at 1 and 2, and $\ell_0(0) = \frac{1}{2}\cdot(-1)(-2) = 1$.
- *Over a finite field.* In $\mathbb{F}_3$, with the nodes $0, 1, 2$, the same formula gives $\ell_0 = 2^{-1}(x-1)(x-2) = 2(x-1)(x-2)$, since $2\cdot 2 = 4 = 1$ in $\mathbb{F}_3$. Its value at 0 is $2\cdot(-1)(-2) = 4 = 1$.

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↻ **Second reading** · *One node, and a repeated node*

*One node* ($n = 0$). The product defining $\ell_0$ has no factors, so $\ell_0 = 1$, the empty product. The theorem below then says that the only polynomial of degree at most 0 with value $y_0$ at $c_0$ is the constant $y_0$, which is right.

*A repeated node.* Change the nodes $0, 1$ of the first example to $0, 0$. The clause that fails is "distinct": the factor $(c_0 - c_1)^{-1} = 0^{-1}$ does not exist. Nothing can repair this. A polynomial cannot have the value 1 at 0 and also the value 2 at 0, so with a repeated node some prescribed values have no interpolating polynomial at all.

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The next theorem says that the Lagrange polynomials do what they were built for, and that the polynomial they produce is the only one.

**Theorem 5.5.1. Lagrange Interpolation**

Let $n \in \mathbb{N}$, let $c_0, \dots, c_n \in F$ be distinct, and let $y_0, \dots, y_n \in F$. Let $\ell_0, \dots, \ell_n$ be the Lagrange polynomials of the nodes (Definition 5.5.1).

(a) For all $i, k \in \{0, \dots, n\}$, $\ell_i(c_k) = 1$ if $k = i$ and $\ell_i(c_k) = 0$ if $k \neq i$. Each $\ell_i$ has degree exactly $n$.

(b) The polynomial $p := \sum_{i=0}^{n} y_i \ell_i$ satisfies $\deg p \leq n$ and $p(c_k) = y_k$ for every $k$.

(c) It is the **only** polynomial of degree at most $n$ with $p(c_k) = y_k$ for every $k$.

(d) Every $q \in F[x]$ with $\deg q \leq n$ satisfies $q = \sum_{i=0}^{n} q(c_i)\ell_i$.

*Try it first (ten minutes): for the nodes $0, 1, 2 \in \mathbb{Q}$, write down $\ell_0, \ell_1, \ell_2$ and check (a) on them. Then find a quadratic through $(0, 1)$, $(1, 3)$, $(2, 7)$. Why can there be no second one?*

*Idea.* The direct attempt writes $p = a_0 + a_1 x + a_2 x^2$ and solves the three equations $p(0) = 1$, $p(1) = 3$, $p(2) = 7$ for $a_0, a_1, a_2$. It works here, but it gives no formula, and for $n + 1$ nodes it is a system of $n + 1$ equations. The Lagrange polynomials replace the system by a sum: (a) is "evaluate at a root", since every factor $x - c_j$ with $j \neq i$ vanishes at $c_j$, and (b) follows from (a) term by term. The second question, why there is no other quadratic, is not answered by any construction. It is answered by counting roots: two such quadratics differ by a polynomial of degree at most 2 with three roots.

*Proof.*

(a) Let $i, k \in \{0, \dots, n\}$. Evaluation respects products (Theorem 4.4.1), so $\ell_i(c_k)$ is the

product of the values of the factors at $c_k$:

$$\ell_i(c_k) = \prod_{\substack{j=0 \\ j \neq i}}^{n} (c_i - c_j)^{-1}(c_k - c_j).$$

If $k \neq i$, the factor with $j = k$ is $(c_i - c_k)^{-1} \cdot 0 = 0$, so the product is 0. If $k = i$, every factor is $(c_i - c_j)^{-1}(c_i - c_j) = 1$, so the product is 1. For the degree, each factor $(c_i - c_j)^{-1}x - (c_i - c_j)^{-1}c_j$ has degree 1, since its coefficient of $x$ is non-zero. A field has no zero divisors, so by Theorem 4.3.2 (b) the degree of a product is the sum of the degrees; with $n$ factors, $\deg \ell_i = n$.

(b) Each term $y_i\ell_i$ has degree at most $n$ (Theorem 4.3.2 (a)), so $\deg p \leq n$ by Theorem 4.3.1, applied one term at a time. Evaluating at $c_k$ and using (a),

$$p(c_k) = \sum_{i=0}^{n} y_i\ell_i(c_k) = y_k,$$

since every term with $i \neq k$ is $y_i \cdot 0 = 0$, and the term $i = k$ is $y_k \cdot 1$ ((Σ3) of Lemma 4.1.1).

(c) Let $q \in F[x]$ with $\deg q \leq n$ and $q(c_k) = y_k$ for every $k$. Then $p$ and $q$ both have degree at most $n$ and agree at the $n + 1$ distinct points $c_0, \ldots, c_n$. By Theorem 5.4.2 (a), $q = p$.

(d) Let $\deg q \leq n$, and apply (b) and (c) with $y_i \coloneqq q(c_i)$. The polynomial $\sum_i q(c_i)\ell_i$ and $q$ itself both have degree at most $n$ and the values $q(c_0), \ldots, q(c_n)$ at the nodes, so they are equal by (c). This proves the theorem.

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So for distinct nodes and any values, there is exactly one polynomial of degree at most $n$ through the points, and (b) writes it down. Part (d) says that a polynomial of degree at most $n$ is determined by its values at $n + 1$ nodes, and rebuilt from them by the same formula. Where the linear-algebra companion, Chapter 6, speaks of "the Lagrange basis", this book has only the formula (d); no basis is defined here.

*Warning.* The bound "degree at most $n$" in (c) is what makes the interpolant unique. Without it, uniqueness fails: $p$ and $p + (x - c_0)(x - c_1)\cdots(x - c_n)$ have the same values at every node, and the second has degree $n + 1$. For the nodes $0, 1$ and the values $5, 8$, both $5 + 3x$ and $5 + 3x + x(x - 1) = 5 + 2x + x^2$ pass through $(0, 5)$ and $(1, 8)$.

*Quick check.* Write down the Lagrange polynomials of the nodes $0, 1$ in $\mathbb{Q}$, and use them to find the line through $(0, 5)$ and $(1, 8)$.

*Solution.* $\ell_0 = 1 - x$ and $\ell_1 = x$ (the first example after Definition 5.5.1). By Theorem 5.5.1 (b), the line is $5\ell_0 + 8\ell_1 = 5(1 - x) + 8x = 5 + 3x$. Check: its values at 0 and 1 are 5 and 8.

Here is a cubic, worked through the formula.

**Example. A cubic through four points**

Find the polynomial $p \in \mathbb{Q}[x]$ of degree at most 3 with $p(-1) = 2$, $p(0) = 1$, $p(1) = 2$ and $p(2) = 11$.

*Solution.* The nodes $c_0 = -1$, $c_1 = 0$, $c_2 = 1$, $c_3 = 2$ are distinct. From Definition 5.5.1, with the denominators computed first,

$$\begin{aligned}
\ell_0 &= ((-1)(-2)(-3))^{-1} x(x-1)(x-2) = (-6)^{-1}(x^3 - 3x^2 + 2x),\\
\ell_1 &= ((1)(-1)(-2))^{-1} (x+1)(x-1)(x-2) = 2^{-1}(x^3 - 2x^2 - x + 2),\\
\ell_2 &= ((2)(1)(-1))^{-1} (x+1)x(x-2) = (-2)^{-1}(x^3 - x^2 - 2x),\\
\ell_3 &= ((3)(2)(1))^{-1} (x+1)x(x-1) = 6^{-1}(x^3 - x).
\end{aligned}$$

By Theorem 5.5.1 (b), $p = 2\ell_0 + \ell_1 + 2\ell_2 + 11\ell_3$. Collect each power of $x$:

$$\begin{aligned}
x^3 &: \ -\tfrac{1}{3} + \tfrac{1}{2} - 1 + \tfrac{11}{6} = 1, \qquad x^2 : \ 1 - 1 + 1 = 1,\\
x &: \ -\tfrac{2}{3} - \tfrac{1}{2} + 2 - \tfrac{11}{6} = -1, \qquad 1 : \ 0 + 1 + 0 + 0 = 1.
\end{aligned}$$

So $p = x^3 + x^2 - x + 1$. Check the four values: $p(-1) = -1 + 1 + 1 + 1 = 2$, $p(0) = 1$, $p(1) = 1+1-1+1 = 2$, $p(2) = 8+4-2+1 = 11$. By Theorem 5.5.1 (c), no other polynomial of degree at most 3 has these values.

## 5.5.2 Congruence modulo a polynomial

The opening question turned values into remainders. To compute with remainders as *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapter 3 did with integers, we need the relation "same remainder" and the classes it makes.

**Definition 5.5.2. Congruence Modulo a Polynomial**

Let $m \in F[x]$. Polynomials $f, g \in F[x]$ are **congruent modulo** $m$, written $f \equiv g \pmod{m}$, if $m \mid f - g$ (Definition 5.1.1), that is, if $f - g \in \langle m \rangle$ (Definition 5.2.1).

In words: $f$ and $g$ differ by a multiple of $m$. The set $\langle m \rangle$ of multiples of $m$ is an ideal of the commutative ring $F[x]$ (Definition 5.2.1), so the relation is the relation $\sim_I$ of Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.1 for the ideal $I = \langle m \rangle$, and Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.1 (a) makes it an equivalence relation. We write $[f] = f + \langle m \rangle$ for the class of $f$, and $[f]_m$ when the modulus must be named.

**Examples.**

- $x^2 \equiv -1 \pmod{x^2 + 1}$, since $x^2 - (-1) = 1 \cdot (x^2 + 1)$.
- For every $c \in F$, $f \equiv f(c) \pmod{x - c}$: by Theorem 5.1.3 (a), $f = q(x - c) + f(c)$ for some $q$, so $f - f(c)$ is a multiple of $x - c$.
- Two polynomials are congruent modulo $m$ exactly when they leave the same remainder on division by $m \neq 0$; this is (b) of the next theorem.

**Second reading** · *The degenerate moduli 0 and the constants*

For $m = 0$, $f \equiv g \pmod 0$ means $0 \mid f - g$, that is, $f = g$ (Proposition 5.1.1 (a)): every class is a single polynomial. For a non-zero constant $m$, every $f$ is a multiple of $m$ (Proposition 5.1.1 (a)), so all polynomials are congruent and there is one class; the ring it forms is the zero ring, in which $[1] = [0]$. Both cases are legal and neither is useful, which is why the next theorem asks for $\deg m \geq 1$.

*Quick check.* Is $x^3 \equiv x \pmod{x^2+1}$ in $\mathbb{Q}[x]$? In $\mathbb{F}_2[x]$?

*Solution.* Divide: $x^3 = x(x^2+1) - x$, so $x^3 \equiv -x$, and $x^3 - x = x(x^2+1) - 2x$. Hence $x^3 \equiv x$ exactly when $x^2+1 \mid -2x$. In $\mathbb{Q}[x]$ the polynomial $-2x$ is non-zero of degree $1 < 2$, so $x^2+1$ does not divide it (Proposition 5.1.1 (d)): **no**. In $\mathbb{F}_2[x]$, $-2x = 0$, which every polynomial divides: **yes**. The same symbols give different answers over different fields.

## 5.5.3 The ring of classes modulo a polynomial

*Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapter 3 built $\mathbb{Z}/n\mathbb{Z}$ as the quotient of $\mathbb{Z}$ by the ideal $n\mathbb{Z}$, and found in each class exactly one remainder in $\{0, \ldots, n-1\}$. The same construction works for $F[x]$ and an ideal $\langle m \rangle$, and division with remainder again picks one representative per class, now "of degree less than $\deg m$".

**Theorem 5.5.2. The Ring of Polynomials Modulo m**

Let $m \in F[x]$ with $d \coloneqq \deg m \geq 1$, and write $[f] = f + \langle m \rangle$.

(a) $F[x]/\langle m \rangle$, with $[f] + [g] \coloneqq [f+g]$ and $[f][g] \coloneqq [fg]$, is a commutative ring with zero $[0]$ and one $[1]$, and $[1] \neq [0]$.

(b) Each class contains **exactly one** polynomial of degree less than $d$: the remainder of any of its members on division by $m$. In particular $f \equiv g \pmod m$ if and only if $f$ and $g$ leave the same remainder, and sending each class to its member of degree less than $d$ is a bijection from $F[x]/\langle m \rangle$ onto $\{r \in F[x] : \deg r < d\}$.

(c) The map $F \to F[x]/\langle m \rangle$, $c \mapsto [c]$ ($c$ a constant), is injective and preserves $+$ and $\cdot$.

(d) $[f]$ is a unit of $F[x]/\langle m \rangle$ (Number Systems: From \(\nN\) to \(\nR\), Definition 3.5.1) **if and only if** $\gcd(f, m) = 1$. In that case $[f]^{-1} = [a]$ for **any** $a, b \in F[x]$ with $af + bm = 1$.

*Idea.* Part (a) is the quotient theorem of *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$ applied to the ideal $\langle m \rangle$; only $[1] \neq [0]$ needs the degree. Part (b) is division with remainder read twice: existence of the remainder puts one polynomial of small degree into each class, and "compare degrees" shows a class cannot hold two, since their difference would be a non-zero multiple of $m$ of

smaller degree than $m$. Part (d) translates $[a][f] = [1]$ into a Bézout identity, exactly as for $\mathbb{Z}/n\mathbb{Z}$.

*Proof.*

(a) By Definition 5.2.1, $\langle m\rangle$ is an ideal of the commutative ring $F[x]$, so Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.1 (a)–(c) give the commutative ring $F[x]/\langle m\rangle$ with these operations, zero [0] and one [1]. By Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.1 (d), $[1] = [0]$ would mean $1 \in \langle m\rangle$, that is, $m \mid 1$; then $\deg m \le \deg 1 = 0$ by Proposition 5.1.1 (d), against $d \ge 1$. So $[1] \neq [0]$.

(b) Since $d \ge 1$, $m \neq 0$. *Existence.* Let $f \in F[x]$. By Theorem 5.1.2 (a), $f = qm + r$ with $\deg r < d$. Then $f - r = qm \in \langle m\rangle$, so $r \in [f]$. *Uniqueness.* Let $r, r' \in [f]$ with $\deg r, \deg r' < d$. Then $r - r' \in \langle m\rangle$ (both are $\sim$ to $f$, and $\sim$ is an equivalence relation), so $m \mid r - r'$, and $\deg(r - r') < d$ by Theorem 4.3.1. If $r - r' \neq 0$, then $d \le \deg(r - r')$ by Proposition 5.1.1 (d), a contradiction. So $r = r'$.
Hence two polynomials are congruent exactly when their classes are equal (Sets, Relations, and Cardinality, Theorem 2.5.1 (b)), exactly when the classes have the same member of degree less than $d$, which is the common remainder. The map "class $\mapsto$ its member of degree less than $d$" is a function by the uniqueness just proved. It is injective, since two classes with a common member are equal (Sets, Relations, and Cardinality, Theorem 2.5.1 (c)); and it is surjective, since a polynomial $r$ with $\deg r < d$ is the member of degree less than $d$ of its own class $[r]$.

(c) For constants $c, c'$, $[c]+[c'] = [c+c']$ and $[c][c'] = [cc']$ by the definition of the operations, where $c + c'$ and $cc'$ are computed in $F$ (Theorem 4.2.4 (b)). If $[c] = [c']$, then $c$ and $c'$ are members of one class of degree at most $0 < d$, so $c = c'$ by (b).

(d) ($\Leftarrow$) Suppose $\gcd(f, m) = 1$. By Corollary 5.2.3 there are $a, b \in F[x]$ with $af + bm = 1$; let $a, b$ be any such pair. Then $af - 1 = -bm \in \langle m\rangle$, so $[a][f] = [af] = [1]$. So $[f]$ is a unit, and its inverse, which is unique (Number Systems: From \(\nN\) to \(\nR\), Definition 3.5.1), is $[a]$.

($\Rightarrow$) Suppose $[a][f] = [1]$ for some $a \in F[x]$ (every element of the quotient is a class, Sets, Relations, and Cardinality, Definition 2.5.4). Then $af - 1 \in \langle m\rangle$: $af - 1 = km$ for some $k \in F[x]$. Hence $af + (-k)m = 1$, and $\gcd(f, m) = 1$ by Corollary 5.2.3. This proves the theorem.

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So $F[x]/\langle m\rangle$ is a ring whose elements may be named by the polynomials of degree less than $\deg m$, and by (c) its constants form a copy of $F$. From now on we write $c$ for the class $[c]$ of a constant when no confusion can arise. A computation in the quotient is: compute with representatives, then replace the result by its remainder. The integer twins are Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (b) (one remainder per class) and Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.1 (units).

*Quick check.* In $\mathbb{Q}[x]/\langle x^2 - 2\rangle$, show that $[x + 1]$ is a unit, and find its inverse with (d).

*Solution.* Divide: $x^2 - 2 = (x-1)(x+1) - 1$. Hence $(x-1)(x+1) + (-1)(x^2-2) = 1$, so $\gcd(x+1, x^2-2) = 1$ (Corollary 5.2.3), and by Theorem 5.5.2 (d) with $a = x-1$, $[x+1]$ is a unit with $[x+1]^{-1} = [x-1]$. Check: $(x+1)(x-1) = x^2 - 1 \equiv 2 - 1 = 1 \pmod{x^2-2}$.

When is every non-zero class a unit? By (d), when every $f$ with $m \nmid f$ is coprime to $m$. For an irreducible $m$, Corollary 5.3.3 (a) says exactly this. For a reducible $m$, a factor of $m$ of smaller positive degree is neither a multiple of $m$ nor coprime to it.

**Theorem 5.5.3. The Quotient Is a Field Exactly When m Is Irreducible**

Let $m \in F[x]$ with $\deg m \geq 1$. Then $F[x]/\langle m \rangle$ is a field (Definition 2.1.1) **if and only if** $m$ is irreducible over $F$ (Definition 5.3.1).

*Proof.* By Theorem 5.5.2 (a), $F[x]/\langle m \rangle$ is a commutative ring with $[1] \neq [0]$. So it is a field exactly when every non-zero element is a unit.

($\Leftarrow$) Let $m$ be irreducible and $[f] \neq [0]$. Then $f \notin \langle m \rangle$ by Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.1 (d), that is, $m \nmid f$. By Corollary 5.3.3 (a), $\gcd(m, f) = 1$; the definition of the gcd (Definition 5.2.2) is symmetric in its two arguments, so $\gcd(f, m) = 1$. Hence $[f]$ is a unit by Theorem 5.5.2 (d).

($\Rightarrow$) Let $m$ be not irreducible. Since $\deg m \geq 1$, Definition 5.3.1 gives $g, h \in F[x]$, neither a constant, with $m = gh$. Then $g, h \neq 0$ and $\deg g, \deg h \geq 1$, and $\deg g + \deg h = \deg m$ (Theorem 4.3.2 (b)), so $\deg g, \deg h < \deg m$. A non-zero polynomial of degree less than $\deg m$ is not a multiple of $m$ (Proposition 5.1.1 (d)), so $[g] \neq [0]$ and $[h] \neq [0]$. But $[g][h] = [gh] = [m] = [0]$. In a field a product of non-zero elements is non-zero (Theorem 2.1.1 (f)), so $F[x]/\langle m \rangle$ is not a field. This proves the theorem.

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This is Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.2 with "prime" replaced by "irreducible". It manufactures new fields: from $F$ and an irreducible $m$ of degree $d$, a field whose elements are named by the polynomials of degree less than $d$.

**Example. A field with four elements, and the complex numbers again**

(i) Describe $\mathbb{F}_2[x]/\langle x^2 + x + 1 \rangle$, and show that it is a field with four elements.
(ii) Describe $\mathbb{R}[x]/\langle x^2 + 1 \rangle$.

*Solution.*

(i) Let $m := x^2 + x + 1 \in \mathbb{F}_2[x]$. Its values are $m(0) = 1$ and $m(1) = 1 + 1 + 1 = 1$ in $\mathbb{F}_2$, so $m$ has no root in $\mathbb{F}_2$; it has degree 2, so it is irreducible by Proposition 5.3.1 (b), and the quotient is a field by Theorem 5.5.3. By Theorem 5.5.2 (b), its elements are the classes of the polynomials of degree less than 2 over $\mathbb{F}_2$, namely $0, 1, x, x+1$: four elements. Write $\alpha := [x]$. Since $x^2 - (x+1) = x^2 + x + 1$ in $\mathbb{F}_2[x]$ (where $-1 = 1$),

$x^2 \equiv x+1$, and so

$$\alpha^2 = \alpha + 1, \qquad \alpha(\alpha+1) = \alpha^2 + \alpha = 2\alpha + 1 = 1, \qquad (\alpha+1)^2 = \alpha^2 + 2\alpha + 1 = \alpha,$$

using $2 = 0$ in $\mathbb{F}_2$. The multiplication table of the non-zero elements is therefore

| $\cdot$ | $1$ | $\alpha$ | $\alpha+1$ |
|---|---|---|---|
| $1$ | $1$ | $\alpha$ | $\alpha+1$ |
| $\alpha$ | $\alpha$ | $\alpha+1$ | $1$ |
| $\alpha+1$ | $\alpha+1$ | $1$ | $\alpha$ |

and every non-zero element has an inverse in it, as the theorem promised. Also $\alpha^3 = \alpha \cdot \alpha^2 = \alpha(\alpha+1) = 1$. This is the four-element field of Exercise (C1: A Field with Four Elements). There the rule $\alpha^2 = 1 + \alpha$ is part of the definition of the multiplication; here it is computed, as the remainder of $x^2$ on division by $x^2 + x + 1$.

(ii) The polynomial $x^2 + 1 \in \mathbb{R}[x]$ has no real root, since $c^2 + 1 \geq 1 > 0$ for every $c \in \mathbb{R}$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (e)); it has degree 2, so it is irreducible (Proposition 5.3.1 (b)), and $\mathbb{R}[x]/\langle x^2+1\rangle$ is a field. Its elements are the classes $[a+bx]$ with $a, b \in \mathbb{R}$, each named by exactly one such polynomial, and $[x]^2 = [x^2] = [-1]$. So it is a field containing a copy of $\mathbb{R}$ (Theorem 5.5.2 (c)) and an element whose square is $-1$: it looks like the complex numbers of Chapter 1, and Exercise B3 shows it is a copy of them.

The theorem needs "irreducible", and having no root is not a substitute: Section 5.3 showed that the two differ from degree 4 on.

*Warning.* "$m$ has no root, so $F[x]/\langle m\rangle$ is a field" is false. The polynomial $m = (x^2+1)^2 \in \mathbb{R}[x]$ has no real root: for every $c \in \mathbb{R}$, $c^2 + 1 > 0$ (Example (A field with four elements, and the complex numbers again) (ii)), so $m(c) = (c^2+1)^2 > 0$ by Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (e). But $[x^2+1] \neq [0]$, since $x^2+1$ has degree $2 < 4 = \deg m$, while $[x^2+1]^2 = [(x^2+1)^2] = [0]$. A non-zero element with square zero cannot be a unit, so $\mathbb{R}[x]/\langle (x^2+1)^2\rangle$ is not a field.

*Quick check.* Is $\mathbb{Q}[x]/\langle x^2-2\rangle$ a field? Is $\mathbb{R}[x]/\langle x^2-2\rangle$?

*Solution.* $\mathbb{Q}[x]/\langle x^2-2\rangle$: **yes**. A rational root $c$ would give $c^2 = 2$, impossible by Number Systems: From \(\nN\) to \(\nR\), Theorem 2.6.1; so $x^2-2$ is irreducible over $\mathbb{Q}$ (Proposition 5.3.1 (b)), and Theorem 5.5.3 applies. $\mathbb{R}[x]/\langle x^2-2\rangle$: **no**. With $s$ the positive square root of 2 in $\mathbb{R}$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 6.1.4), $x^2 - 2 = (x-s)(x+s)$ is reducible over $\mathbb{R}$, and the theorem says the quotient is not a field; concretely $[x-s][x+s] = [0]$.

## 5.5.4 The Chinese remainder theorem for polynomials

Now several congruences at once. *Number Systems: From $\mathbb{N}$ to $\mathbb{R}$*, Chapter 3 solved two congruences to coprime integer moduli by writing 1 as a combination of the moduli and multiplying (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.3). With polynomials the move is the same, and we take $k$ moduli at once. The one new ingredient is that coprimality survives products.

**Lemma 5.5.4. Coprime to Each Factor, Coprime to the Product**

(a) Let $s \geq 1$ and $p, q_1, \ldots, q_s \in F[x]$. If $\gcd(p, q_j) = 1$ for every $j$, then $\gcd(p, q_1 \cdots q_s) = 1$.

(b) Let $k \geq 1$ and let $m_1, \ldots, m_k \in F[x]$ be **pairwise coprime**: $\gcd(m_i, m_j) = 1$ whenever $i \neq j$. If $m_i \mid h$ for every $i$, then $m_1 \cdots m_k \mid h$.

*In words: a polynomial coprime to each factor is coprime to the product, and pairwise coprime divisors of $h$ divide $h$ together.*

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↻ **Second reading** · *The proof: multiply the Bézout identities*

*Idea.* For (a), write 1 as a combination of $p$ and each factor, and multiply the identities: every term of the product but one contains $p$, and the remaining term is a multiple of the product of the factors. For (b), the product of the first $k$ moduli is coprime to the next one by (a), so two coprime divisors combine (Proposition 5.3.4); induction adds one modulus at a time.

*Proof.*

(a) We apply Logic and Proof, Axiom 4.2.1 (on $\mathbb{N} = \omega$, Number Systems: From \(\nN\) to \(\nR\), Corollary 1.7.3 (b)) to $P(n)$: "the claim holds for $s = n + 1$". $P(0)$, the case $s = 1$, has nothing to prove. Suppose $P(n)$, that is, the claim for $s = n + 1$, and let $p$ be coprime to each of $q_1, \ldots, q_{s+1}$. Write $Q \coloneqq q_1 \cdots q_s$, so that $q_1 \cdots q_{s+1} = Qq_{s+1}$ by the product recursion (Sets, Relations, and Cardinality, Definition 2.3.1, read in $F[x]$). By the induction hypothesis and Corollary 5.2.3, $ap + bQ = 1$ and $a'p + b'q_{s+1} = 1$ for some $a, b, a', b' \in F[x]$. Multiplying the two identities and expanding,

$$p\,(aa'p + ab'q_{s+1} + a'bQ) + (bb')\,Qq_{s+1} = 1.$$

So $\gcd(p, Qq_{s+1}) = 1$ by Corollary 5.2.3.

(b) We apply Logic and Proof, Axiom 4.2.1 (on $\mathbb{N} = \omega$, Number Systems: From \(\nN\) to \(\nR\), Corollary 1.7.3 (b)) to $P(n)$: "the claim holds for $k = n + 1$". $P(0)$, the case $k = 1$, has nothing to prove. Suppose $P(n)$, that is, the claim for $k = n + 1$, and let $m_1, \ldots, m_{k+1}$ be pairwise coprime, each dividing $h$. With $M \coloneqq m_1 \cdots m_k$, the induction hypothesis gives $M \mid h$. The modulus $m_{k+1}$ is coprime to each of $m_1, \ldots, m_k$, so $\gcd(m_{k+1}, M) = 1$ by (a), and $\gcd(M, m_{k+1}) = 1$ since the gcd is symmetric (Definition 5.2.2). By Proposition 5.3.4, $Mm_{k+1} = m_1 \cdots m_{k+1}$ divides $h$.

■

Without coprimality (b) fails: for $m_1 = m_2 = x$ and $h = x$, each $m_i$ divides $h$, but $m_1 m_2 = x^2$ does not (Proposition 5.1.1 (d)); here $\gcd(m_1, m_2) = x \neq 1$.

**Theorem 5.5.5. Chinese Remainder Theorem for Polynomials**

Let $k \geq 1$, let $m_1, \ldots, m_k \in F[x]$ be non-zero and pairwise coprime, and let $m := m_1 \cdots m_k$. For each $i$, let $M_i$ be the product of the $m_j$ with $j \neq i$ (in increasing order of $j$).

(a) For each $i$ there are $a_i, b_i \in F[x]$ with $a_i M_i + b_i m_i = 1$. Put $e_i := a_i M_i$. Then $e_i \equiv 1 \pmod{m_i}$ and $e_i \equiv 0 \pmod{m_j}$ for $j \neq i$, and for all $r_1, \ldots, r_k \in F[x]$ the polynomial $f := \sum_{i=1}^{k} r_i e_i$ satisfies

$$f \equiv r_i \pmod{m_i} \qquad \text{for every } i \in \{1, \ldots, k\}.$$

(b) A polynomial $g$ satisfies the same $k$ congruences **if and only if** $g \equiv f \pmod{m}$.

(c) Exactly one polynomial of degree less than $\deg m$ satisfies the $k$ congruences.

(d) The rule $\Phi([f]_m) := ([f]_{m_1}, \ldots, [f]_{m_k})$ is a well-defined **bijection**

$$\Phi \colon F[x]/\langle m \rangle \to F[x]/\langle m_1 \rangle \times \cdots \times F[x]/\langle m_k \rangle,$$

and, with $+$ and $\cdot$ on the right taken componentwise, $\Phi$ preserves $+$ and $\cdot$ and sends $[1]_m$ to $([1]_{m_1}, \ldots, [1]_{m_k})$.

*Idea.* This is the integer proof of Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.3 with $k$ moduli. The identity $a_i M_i + b_i m_i = 1$ makes a polynomial $e_i$ that is 1 modulo $m_i$ and 0 modulo every other modulus, as $\ell_i$ is 1 at one node and 0 at the others. Then $\sum r_i e_i$ has every remainder right. Uniqueness modulo $m$ is Lemma 5.5.4 (b): a difference divisible by every $m_i$ is divisible by their product. Part (c) picks the remainder modulo $m$, and (d) restates (a) and (b) as a bijection.

*Proof.* By (L2) of Section 5.3, applied to the list $(m_1, \ldots, m_k)$ with the entry $i$ removed, $m = M_i m_i$, so $m_i \mid m$; and $m_j \mid M_i$ for $j \neq i$, by the last clause of (L2), since $m_j$ is an entry of the list whose product is $M_i$. By (L1) of Section 5.3, $m \neq 0$, since every $m_i$ is non-zero.

(a) If $k = 1$, then $M_1 = 1$ (the empty product), and $a_1 := 1$, $b_1 := 0$ will do. If $k \geq 2$, then $m_i$ is coprime to each $m_j$ with $j \neq i$, so Lemma 5.5.4 (a) gives $\gcd(m_i, M_i) = 1$; by the symmetry of the gcd and Corollary 5.2.3, $a_i M_i + b_i m_i = 1$ for some $a_i, b_i$. In either case $e_i - 1 = -b_i m_i$, so $e_i \equiv 1 \pmod{m_i}$; and for $j \neq i$, $m_j \mid M_i \mid e_i$, so $e_i \equiv 0 \pmod{m_j}$.
Now fix $j$. In $f - r_j = \sum_{i \neq j} r_i e_i + r_j(e_j - 1)$, every term with $i \neq j$ is a multiple of $m_j$, and so is $r_j(e_j - 1) = -r_j b_j m_j$. A sum of multiples of $m_j$ is a multiple of $m_j$ (Proposition 5.1.1 (c), and induction on the number of terms). So $f \equiv r_j \pmod{m_j}$.

(b) ($\Rightarrow$) Suppose $g \equiv r_i \pmod{m_i}$ for every $i$. Then $m_i \mid (g - r_i) - (f - r_i) = g - f$ for every $i$ (Proposition 5.1.1 (c)). The $m_i$ are pairwise coprime, so $m \mid g - f$ by Lemma 5.5.4 (b). ($\Leftarrow$) Suppose $m \mid g - f$. Since $m_i \mid m$, $m_i \mid g - f$ (Proposition 5.1.1 (b)), and then

$m_i \mid (g-f)+(f-r_i) = g-r_i$ for every $i$.

(c) *Existence.* Since $m \neq 0$, divide $f$ by $m$: $f = qm+r$ with $\deg r < \deg m$ (Theorem 5.1.2). Then $r \equiv f \pmod m$, so $r$ satisfies the $k$ congruences by (b). *Uniqueness.* If $g$ and $g'$ both satisfy them and have degree less than $\deg m$, then $g \equiv f \equiv g' \pmod m$ by (b), so $m \mid g-g'$, while $\deg(g-g') < \deg m$ (Theorem 4.3.1). A non-zero multiple of $m$ has degree at least $\deg m$ (Proposition 5.1.1 (d)), so $g = g'$. (If every $m_i$ is a non-zero constant, then $\deg m = 0$, and the one solution of degree less than 0 is the zero polynomial.)

(d) *Well defined.* Fix $i$. If $f \equiv g \pmod m$, then $f \equiv g \pmod{m_i}$, since $m_i \mid m$. By Number Systems: From \(\nN\) to \(\nR\), Theorem 2.2.1 (a), applied to the identity map of $F[x]$ with congruence modulo $m$ on the source and modulo $m_i$ on the target, there is exactly one function $F[x]/\langle m\rangle \to F[x]/\langle m_i\rangle$ sending $[f]_m$ to $[f]_{m_i}$. Taking these $k$ functions together gives $\Phi$.

*Surjective.* An element of the target is $([r_1]_{m_1}, \dots, [r_k]_{m_k})$ for some $r_1, \dots, r_k \in F[x]$ (Sets, Relations, and Cardinality, Definition 2.5.4). With $f$ as in (a), $[f]_{m_i} = [r_i]_{m_i}$ for every $i$ (Sets, Relations, and Cardinality, Theorem 2.5.1 (b)), so $\Phi([f]_m)$ is that element.

*Injective.* If $\Phi([f]_m) = \Phi([g]_m)$, then $[f]_{m_i} = [g]_{m_i}$, that is, $m_i \mid g-f$, for every $i$ (Sets, Relations, and Cardinality, Theorem 2.5.1 (b)). By Lemma 5.5.4 (b), $m \mid g-f$, that is, $[g]_m = [f]_m$.

*Operations.* By the definition of $+$ in each quotient (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.1 (b)) and of the componentwise sum,

$$\begin{aligned}\Phi([f]_m + [g]_m) &= \Phi([f+g]_m) = ([f+g]_{m_1}, \dots, [f+g]_{m_k}) \\ &= ([f]_{m_1} + [g]_{m_1}, \dots, [f]_{m_k} + [g]_{m_k}) = \Phi([f]_m) + \Phi([g]_m).\end{aligned}$$

The same lines with $fg$ in place of $f+g$ give the product, and $\Phi([1]_m) = ([1]_{m_1}, \dots, [1]_{m_k})$ by definition. This proves the theorem.

■

In (a) the solution is built from the moduli alone, and the right-hand sides $r_i$ enter linearly; the $e_i$ depend only on $m_1, \dots, m_k$. Part (c) is the version used in computation: reduce any solution modulo $m$. Compared with Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.3, this theorem allows any number $k \geq 1$ of moduli in one step, where the integer theorem combines two at a time.

*Quick check.* Find the polynomial $f \in \mathbb{Q}[x]$ of degree less than 2 with $f \equiv 3 \pmod x$ and $f \equiv 5 \pmod{x-1}$.

*Solution.* By Theorem 5.1.3 (b), $f \equiv 3 \pmod x$ says that 0 is a root of $f-3$, that is, $f(0) = 3$; likewise the second congruence says $f(1) = 5$. By Theorem 5.5.1, $f = 3(1-x)+5x = 3+2x$. (The Chinese remainder route: $x-(x-1) = 1$, so $e_1 = -(x-1) = 1-x$ and $e_2 = x$, and $f = 3e_1 + 5e_2$, the same polynomial.)

The check is no accident. Interpolation is the Chinese remainder theorem for the moduli $x - c_i$.

**Proposition 5.5.6. Interpolation Is the Chinese Remainder Theorem for Linear Moduli**

Let $n \in \mathbb{N}$, let $c_0, \dots, c_n \in F$ be distinct, let $y_0, \dots, y_n \in F$, and let $m \coloneqq (x - c_0) \cdots (x - c_n)$.

(a) The moduli $x - c_0, \dots, x - c_n$ are pairwise coprime, and $\deg m = n + 1$.

(b) For $f \in F[x]$ and each $i$: $f \equiv y_i \pmod{x - c_i}$ **if and only if** $f(c_i) = y_i$.

(c) The polynomial of degree less than $n + 1$ given by Theorem 5.5.5 (c) for the right-hand sides $y_0, \dots, y_n$ is $\sum_i y_i \ell_i$, the interpolating polynomial of Theorem 5.5.1. In particular, the remainder of the polynomial $e_i$ of Theorem 5.5.5 (a) on division by $m$ is the Lagrange polynomial $\ell_i$.

*In words: the congruences modulo $x - c_i$ are the values at $c_i$, so the polynomial of the Chinese remainder theorem is the interpolating polynomial, and its building blocks $e_i$, reduced modulo $m$, are the Lagrange polynomials.*

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**Second reading** · *The proof: evaluate, and compare by uniqueness*

*Idea.* Two linear moduli differ by a non-zero constant, which is a Bézout identity after scaling. A congruence modulo $x - c$ is a statement about the value at $c$, by the factor theorem. And two polynomials of degree less than $n + 1$ solving the same congruences are equal, by the uniqueness half of the Chinese remainder theorem; so it is enough to check that the Lagrange formula solves them.

*Proof.*

(a) For $i \neq j$, $(x - c_i) - (x - c_j) = c_j - c_i$, a non-zero constant. Multiplying by its inverse,

$$(c_j - c_i)^{-1}(x - c_i) - (c_j - c_i)^{-1}(x - c_j) = 1,$$

so $\gcd(x - c_i, x - c_j) = 1$ by Corollary 5.2.3. Each factor of $m$ has degree 1, so $\deg m = n + 1$ by Theorem 4.3.2 (b).

(b) By Theorem 5.1.3 (b), $x - c_i \mid f - y_i$ if and only if $c_i$ is a root of $f - y_i$, and $(f - y_i)(c_i) = f(c_i) - y_i$ by Theorem 4.4.1.

(c) The polynomial $p \coloneqq \sum_i y_i \ell_i$ has degree at most $n < n + 1$ and $p(c_i) = y_i$ for every $i$ (Theorem 5.5.1 (b)), so by (b) it satisfies the congruences $p \equiv y_i \pmod{x - c_i}$. By the uniqueness in Theorem 5.5.5 (c), it is the solution of degree less than $\deg m$. For the second claim, fix $i$ and take the right-hand sides 1 at the index $i$ and 0 elsewhere. By Theorem 5.5.5 (a), $e_i$ satisfies these congruences, and so does its remainder on division by $m$, which is congruent to $e_i$ modulo $m$ (Theorem 5.5.5 (b)) and has degree less than $\deg m$. By the first claim, with these right-hand sides, that remainder is $\sum_j y_j \ell_j = \ell_i$.

■

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So two routes lead to the same polynomial. The Lagrange route needs only evaluation and the root count; the Chinese remainder route needs Bézout identities, and in exchange it works for

moduli of any degree. The next two examples use moduli that are not all linear.

**Example. A linear and a quadratic modulus**

Find the polynomial $f \in \mathbb{Q}[x]$ of degree less than 3 with

$$f \equiv 1 \pmod{x-2} \qquad \text{and} \qquad f \equiv 2x-1 \pmod{x^2+1}.$$

*Solution. A Bézout identity.* Divide: $x^2+1 = (x+2)(x-2)+5$. Hence

$$1 = \tfrac{1}{5}(x^2+1) - \tfrac{1}{5}(x+2)(x-2),$$

so the moduli $m_1 := x-2$ and $m_2 := x^2+1$ are coprime (Corollary 5.2.3), and $m = (x-2)(x^2+1) = x^3-2x^2+x-2$ has degree 3.

*The polynomials $e_1, e_2$.* Here $M_1 = x^2+1$ and $M_2 = x-2$. The identity reads $a_1M_1+b_1m_1 = 1$ with $a_1 = \frac{1}{5}$, and $a_2M_2 + b_2m_2 = 1$ with $a_2 = -\frac{1}{5}(x+2)$. So

$$e_1 = \tfrac{1}{5}(x^2+1), \qquad e_2 = -\tfrac{1}{5}(x+2)(x-2) = -\tfrac{1}{5}(x^2-4).$$

*Assemble and reduce.* By Theorem 5.5.5 (a),

$$\begin{aligned} 1\cdot e_1 + (2x-1)e_2 &= \tfrac{1}{5}(x^2+1-(2x-1)(x^2-4)) \\ &= \tfrac{1}{5}(-2x^3+2x^2+8x-3). \end{aligned}$$

Its remainder on division by $m$ is found by adding $\frac{2}{5}m$: since $-2x^3+2x^2+8x-3+2(x^3-2x^2+x-2) = -2x^2+10x-7$,

$$f = -\tfrac{2}{5}x^2 + 2x - \tfrac{7}{5},$$

the unique solution of degree less than 3 (Theorem 5.5.5 (c)).

*Check.* $f(2) = -\frac{8}{5}+4-\frac{7}{5} = 1$, which is the first congruence (Proposition 5.5.6 (b)). Modulo $x^2+1$, replace $x^2$ by $-1$: $f \equiv \frac{2}{5}+2x-\frac{7}{5} = 2x-1$.

Squares of linear moduli prescribe a value and a derivative at once.

**Example. Values and derivatives at once**

(a) Let $c, a, b \in F$ and $f \in F[x]$. Show that $f \equiv a+b(x-c) \pmod{(x-c)^2}$ if and only if $f(c) = a$ and $f'(c) = b$ (the formal derivative of Definition 4.5.1).

(b) Find the polynomial $p \in \mathbb{Q}[x]$ of degree at most 3 with $p(0) = 2$, $p'(0) = -1$, $p(1) = 1$ and $p'(1) = 3$.

*Solution.*

(a) Divide by $(x-c)^2$, which has degree 2: $f = q(x-c)^2+r$ with $\deg r \le 1$ (Theorem 5.1.2). Every polynomial of degree at most 1 can be written $\alpha+\beta(x-c)$ with $\alpha, \beta \in F$: if $r = r_0 + r_1x$, take $\beta := r_1$ and $\alpha := r_0 + r_1c$. Evaluating at $c$ (Theorem 4.4.1),

$f(c) = q(c) \cdot 0 + \alpha = \alpha$. By Theorem 4.5.1 (a)–(c),

$$f' = q'(x-c)^2 + (2 \cdot 1)\, q\,(x-c) + \beta,$$

so $f'(c) = \beta$. Now $f \equiv a + b(x-c)$ if and only if $r \equiv a + b(x-c)$, and both have degree less than 2, so by Theorem 5.5.2 (b) this holds if and only if $\alpha + \beta(x-c) = a + b(x-c)$. Comparing the coefficients of $x$ and then the constant terms, that is $\beta = b$ and $\alpha = a$, that is, $f'(c) = b$ and $f(c) = a$.

(b) By (a), the conditions say $p \equiv 2 - x \pmod{x^2}$ and $p \equiv 1 + 3(x-1) = 3x - 2 \pmod{(x-1)^2}$. The moduli are coprime: expanding,

$$(2x+1)(x-1)^2 + (3-2x)x^2 = (2x^3 - 3x^2 + 1) + (3x^2 - 2x^3) = 1.$$

So, with $m_1 = x^2$ and $m_2 = (x-1)^2$, Theorem 5.5.5 (a) gives $e_1 = 2x^3 - 3x^2 + 1$ and $e_2 = 3x^2 - 2x^3$, and

$$(2-x)e_1 + (3x-2)e_2 = -8x^4 + 20x^3 - 12x^2 - x + 2.$$

Adding $8m = 8(x^4 - 2x^3 + x^2)$ gives the remainder modulo $m = x^2(x-1)^2$:

$$p = 4x^3 - 4x^2 - x + 2.$$

Check: $p(0) = 2$; $p' = 12x^2 - 8x - 1$, so $p'(0) = -1$; $p(1) = 4 - 4 - 1 + 2 = 1$; $p'(1) = 12 - 8 - 1 = 3$. By Theorem 5.5.5 (c), $p$ is the only polynomial of degree less than 4 with these four conditions.

This section met the Chinese remainder theorem three times: as interpolation (moduli $x - c_i$), as a value and a derivative at a point (moduli $(x-c)^2$), and in general. In each case the ring $F[x]/\langle m \rangle$ of Theorem 5.5.2 is the place where the computation happens.

## 5.5.5 Exercises

### 5.5.5.1 A. Check your understanding

**Exercise. A1**

(a) Finish the Try it first before Theorem 5.5.1: find the polynomial of degree at most 2 through $(0,1)$, $(1,3)$, $(2,7)$ in $\mathbb{Q}[x]$.

(b) Describe the classes of $F[x]/\langle x \rangle$. Which ring is it, up to renaming?

(c) True or false: $\mathbb{R}[x]/\langle x^2 - 1 \rangle$ is a field. Justify your answer.

(d) Where does the proof of Theorem 5.5.1 (c) use that the nodes are distinct, and where do (a) and (b) use it?

*Solution.*

(a) With the nodes $0, 1, 2$, Definition 5.5.1 gives

$$\ell_0 = \tfrac{1}{2}(x-1)(x-2), \qquad \ell_1 = -x(x-2), \qquad \ell_2 = \tfrac{1}{2}x(x-1).$$

So $p = \ell_0 + 3\ell_1 + 7\ell_2 = \frac{1}{2}(x^2 - 3x + 2) - 3(x^2 - 2x) + \frac{7}{2}(x^2 - x) = x^2 + x + 1$. Check: $p(0) = 1$, $p(1) = 3$, $p(2) = 7$. It is the only one of degree at most 2 (Theorem 5.5.1

(c)).

(b) By Theorem 5.5.2 (b) with $d = 1$, each class contains exactly one polynomial of degree less than 1, a constant; the constant in $[f]$ is $f(0)$, the remainder on division by $x - 0$ (Theorem 5.1.3 (a)). So the classes are the $[c]$, $c \in F$, and they are distinct for distinct $c$. By Theorem 5.5.2 (c), $c \mapsto [c]$ preserves $+$ and $\cdot$; it is injective, and surjective by what was just said. So $F[x]/\langle x\rangle$ is $F$ with each element $c$ renamed $[c]$; and $[f] = [f(0)]$, so passing to the quotient is "evaluate at 0".

(c) **False.** $x^2 - 1 = (x-1)(x+1)$ is reducible, so Theorem 5.5.3 says the quotient is not a field. Concretely, $[x-1] \neq [0] \neq [x+1]$ (degree $1 < 2$), while $[x-1][x+1] = [x^2-1] = [0]$.

(d) Part (c) uses distinctness only through Theorem 5.4.2 (a), which needs the $n+1$ points to be distinct, ultimately because the root count of Section 5.4 counts distinct roots. Parts (a) and (b) use it to form the inverses $(c_i - c_j)^{-1}$ in the definition of $\ell_i$.

#### 5.5.5.2 B. Practice

**Exercise. B1**

Find the polynomial $f \in \mathbb{Q}[x]$ of degree less than 4 with $f \equiv 1 \pmod{x^2}$ and $f \equiv -1 \pmod{x^2+1}$.

*Solution.* The moduli are coprime, since $(x^2+1) - x^2 = 1$ (Corollary 5.2.3), so by Theorem 5.5.5 (c) there is exactly one such $f$. The first congruence says $f - 1 = x^2 h$ for some $h \in \mathbb{Q}[x]$. Then

$$f + 1 = 2 + x^2 h = (2 - h) + h(x^2 + 1),$$

so the second congruence holds if and only if $x^2+1 \mid 2-h$, that is, $h \equiv 2 \pmod{x^2+1}$. The choice $h = 2$ gives $f = 1 + 2x^2$, of degree $2 < 4$. Check: $f - 1 = 2x^2$ and $f + 1 = 2(x^2+1)$. Hence $f = 1 + 2x^2$.

**Exercise. B2**

In $\mathbb{Q}[x]/\langle x^3 - 2\rangle$, show that $[x^2+1]$ is a unit, and compute its inverse.

*Solution.* Run the Euclidean algorithm (Theorem 5.2.5) on $x^3 - 2$ and $x^2 + 1$:

$$x^3 - 2 = x(x^2+1) + (-x-2),$$
$$x^2 + 1 = (-x+2)(-x-2) + 5.$$

The last remainder is the non-zero constant 5, so the gcd is 1. Back-substitute:

$$\begin{aligned} 5 &= (x^2+1) - (-x+2)\big((x^3-2) - x(x^2+1)\big) \\ &= (-x^2+2x+1)(x^2+1) + (x-2)(x^3-2). \end{aligned}$$

Dividing by 5: $\frac{1}{5}(-x^2+2x+1)\cdot(x^2+1) + \frac{1}{5}(x-2)\cdot(x^3-2) = 1$. By Theorem 5.5.2 (d), $[x^2+1]$ is a unit and

$$[x^2+1]^{-1} = \left[\tfrac{1}{5}(-x^2+2x+1)\right].$$

Check: $(x^2+1)(-x^2+2x+1) = -x^4+2x^3+2x+1$, and with $x^3 \equiv 2$, $x^4 \equiv 2x$, this is $\equiv -2x+4+2x+1 = 5$.

**Exercise. B3**

(a) Write the addition and multiplication tables of $\mathbb{F}_2[x]/\langle x^2+x+1\rangle$, with its elements named $0, 1, \alpha, \alpha+1$ as in Example (A field with four elements, and the complex numbers again), and check from the table that every non-zero element is a unit.

(b) Show that $\varphi(a+bi) := [a+bx]$ $(a, b \in \mathbb{R})$ is a bijection $\mathbb{C} \to \mathbb{R}[x]/\langle x^2+1\rangle$ that preserves $+$ and $\cdot$ and sends 1 to $[1]$.

*Solution.*

(a) Addition is coefficientwise in $\mathbb{F}_2$, where $1+1=0$:

| $+$ | $0$ | $1$ | $\alpha$ | $\alpha+1$ |
|---|---|---|---|---|
| $0$ | $0$ | $1$ | $\alpha$ | $\alpha+1$ |
| $1$ | $1$ | $0$ | $\alpha+1$ | $\alpha$ |
| $\alpha$ | $\alpha$ | $\alpha+1$ | $0$ | $1$ |
| $\alpha+1$ | $\alpha+1$ | $\alpha$ | $1$ | $0$ |

| $\cdot$ | $0$ | $1$ | $\alpha$ | $\alpha+1$ |
|---|---|---|---|---|
| $0$ | $0$ | $0$ | $0$ | $0$ |
| $1$ | $0$ | $1$ | $\alpha$ | $\alpha+1$ |
| $\alpha$ | $0$ | $\alpha$ | $\alpha+1$ | $1$ |
| $\alpha+1$ | $0$ | $\alpha+1$ | $1$ | $\alpha$ |

The products come from $\alpha^2 = \alpha+1$, $\alpha(\alpha+1) = 1$, $(\alpha+1)^2 = \alpha$ (Example (A field with four elements, and the complex numbers again)). Each non-zero row of the multiplication table contains 1: $1^{-1}=1$, $\alpha^{-1}=\alpha+1$, $(\alpha+1)^{-1}=\alpha$.

(b) Every complex number is $a+bi$ for exactly one pair $(a,b) \in \mathbb{R}^2$ (Proposition 1.1.1 and the comparison of real and imaginary parts after it), and every element of $\mathbb{R}[x]/\langle x^2+1\rangle$ is $[a+bx]$ for exactly one such pair, by Theorem 5.5.2 (b) (the polynomials of degree less than 2 are the $a+bx$, distinct for distinct pairs). So $\varphi$ is a bijection. Sums: $\varphi$ of $(a+bi)+(c+di) = (a+c)+(b+d)i$ is $[(a+c)+(b+d)x] = [a+bx]+[c+dx]$. Products: in $\mathbb{R}[x]$,

$$(a+bx)(c+dx) = (ac-bd)+(ad+bc)x+bd(x^2+1),$$

so $[a+bx][c+dx] = [(ac-bd)+(ad+bc)x]$, which is $\varphi$ of $(a+bi)(c+di) = (ac-bd)+(ad+bc)i$ (Definition 1.1.1, Proposition 1.1.1). Finally $\varphi(1) = [1]$. So the field of Chapter 1 could have been built as this quotient: the multiplication rule of Definition 1.1.1 is "multiply, then replace $x^2$ by $-1$".

### 5.5.5.3 C. Going deeper

**Exercise. C1: Two routes to the same theorem**

Let $c_0, \dots, c_n \in F$ be distinct and $m := (x-c_0)\cdots(x-c_n)$.

(a) Prove, using Theorem 5.5.1 and the root count but no Bézout identity, that for all $r_0, \dots, r_n \in F[x]$ some $f$ satisfies $f \equiv r_i \pmod{x-c_i}$ for every $i$, and that $g$ satisfies the same congruences if and only if $g \equiv f \pmod{m}$.

(b) Conversely, derive from Theorem 5.5.5 that for all $y_0, \dots, y_n \in F$ exactly one polyno-

mial of degree at most $n$ satisfies $p(c_k) = y_k$ for every $k$ (the content of Theorem 5.5.1 (b) and (c), without the formula $\sum_i y_i \ell_i$).

(c) Which derivation is shorter, and why?

*Solution.*

(a) By the second example after Definition 5.5.2, $r_i \equiv r_i(c_i) \pmod{x - c_i}$. Let $f := \sum_i r_i(c_i)\ell_i$. By Theorem 5.5.1 (b), $f(c_i) = r_i(c_i)$, so $f \equiv r_i(c_i) \equiv r_i \pmod{x - c_i}$, by Proposition 5.5.6 (b) and transitivity. If $g$ also satisfies the congruences, then $g - f \equiv 0 \pmod{x - c_i}$ for every $i$, so each $c_i$ is a root of $g - f$ (Theorem 5.1.3 (b)). The $c_i$ are distinct roots, so $m \mid g - f$ by Theorem 5.4.1. Conversely, if $m \mid g - f$, then each $x - c_i$ divides $m$ (it is an entry of the list whose product is $m$, (L2) of Section 5.3) and so divides $g - f$ (Proposition 5.1.1 (b)), and $g \equiv f \equiv r_i \pmod{x - c_i}$.

(b) By Proposition 5.5.6 (a), (b), the moduli $x - c_i$ are pairwise coprime, $\deg m = n + 1$, and "$f(c_i) = y_i$ for every $i$" is the system $f \equiv y_i \pmod{x - c_i}$. By Theorem 5.5.5 (c), exactly one polynomial of degree less than $n + 1$, that is, of degree at most $n$, solves it. That is the existence of an interpolating polynomial and its uniqueness at once, without the formula for $\ell_i$; identifying it with $\sum_i y_i \ell_i$ still needs Theorem 5.5.1 (a).

(c) The route in (a) is shorter in what it needs: evaluation and the root count. The route in (b) passes through Lemma 5.5.4 and Bézout identities for the moduli. The Lagrange route is shorter because linear moduli are special: "divisible by $x - c$" can be tested by evaluating at $c$, which turns every congruence into an equation in $F$. For moduli of higher degree, evaluation at a single point is no longer enough. For $(x - c)^2$, Example (Values and derivatives at once) (a) needs the derivative as well as the value. For $x^2 + 1$ over $\mathbb{Q}$ in Example (A linear and a quadratic modulus), the modulus has no root in $\mathbb{Q}$, so no evaluation in $\mathbb{Q}$ detects divisibility at all. The Chinese remainder route handles every modulus in the same way.

**Exercise. C2: Every function on a prime field is a polynomial**

Let $p$ be a prime. Prove that every function $\varphi\colon \mathbb{F}_p \to \mathbb{F}_p$ is the polynomial function $c \mapsto f(c)$ of **exactly one** $f \in \mathbb{F}_p[x]$ with $\deg f < p$.

*Solution.* The elements $0, 1, \ldots, p - 1$ of $\mathbb{F}_p$ are $p$ distinct elements (Definition 2.2.1), and there are no others. Use them as nodes, $c_i := i$ for $i = 0, \ldots, p - 1$, so $n = p - 1$.

*Existence.* With $y_i := \varphi(i)$, Theorem 5.5.1 (b) gives $f$ with $\deg f \leq p - 1 < p$ and $f(i) = \varphi(i)$ for every $i$. Since the $i$ are all the elements of $\mathbb{F}_p$, the polynomial function of $f$ is $\varphi$.

*Uniqueness.* If $g$ also has degree less than $p$ and the function $\varphi$, then $f$ and $g$ have degree at most $p - 1$ and agree at the $p$ distinct nodes, so $f = g$ by Theorem 5.5.1 (c).

So over $\mathbb{F}_p$ every function is a polynomial function, and the polynomials of degree less than $p$ correspond one to one to the functions $\mathbb{F}_p \to \mathbb{F}_p$. (As a count: both sets have $p^p$ elements, $p$ choices for each of $p$ coefficients, or for each of $p$ values.) Interpolation over $\mathbb{F}_p$ can use at most $p$ nodes, since $\mathbb{F}_p$ has no more distinct elements. Polynomials of degree $p$ or more are no longer determined by their functions: $x^p$ and $x$ have the same function by Theorem 2.6.6 (a), which is the warning of Section 5.4.